

Classroom Technology Support
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Image Backward process Pure noise

$$q(X_0, X_1, \dots, X_T) = q(X_0) \prod_{t=1}^T q(X_t | X_{t-1})$$

$$q(X_t | X_{t-1}) = \mathcal{N}(X_t; \sqrt{1 - \beta_t} X_{t-1}, \beta_t I)$$

$$q(X_0) = \frac{1}{N}$$

$$q(X_t | X_0) = \mathcal{N}(X_t; \sqrt{\bar{\alpha}_t} X_0, (1 - \bar{\alpha}_t) I)$$

$$\bar{\alpha}_t = \sum_{s=1}^t (1 - \beta_s)$$

$0 < \beta_t < 1$ noise schedule Composition of Gaussians

$$\lim_{T \rightarrow \infty} q(X_T | X_0) = \mathcal{N}(\vec{0}, I)$$

UC San Diego
INFORMATION TECHNOLOGY SERVICES

Diffusion models

Text $\rightarrow$ LLM

Image
Video
Audio

$\rightarrow$ Diffusion models

VAE

$z \rightarrow x$

$z \sim \mathcal{N}(\vec{0}, I)$

$z_0 \sim \mathcal{N}(\vec{0}, I)$

$\downarrow$ p network

image x

Pure noise

$\downarrow$

Image

Denoising diffusion models

$x_T \rightarrow x_{T-1} \rightarrow x_{T-2} \rightarrow \dots \rightarrow x_0$

Pure noise forward process Data generation process

Image

$$P(X_T, X_{T-1}, \dots, X_0) \\ = P(X_T) \prod_{t=1}^T P(X_{t-1} | X_t)$$

$$p(X_T) = N(0, I)$$

$$P(X_{t-1} | X_t) = \frac{P(X_{t-1}, X_t)}{P(X_t)}$$

$$= \frac{p(x_t | x_{t-1}) p(x_{t-1})}{p(x_t)}$$

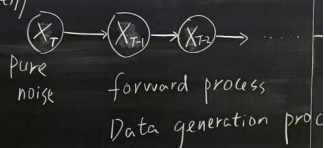
$$p(x_t) = \int_{x_0} p(x_t | x_0) \underbrace{p(x_0)}_{\text{Image}} dx_0$$

$$P(X_{t-1} | X_t)$$

$$p(x_{t-1} | x_t) \approx \mathcal{N}(x_{t-1} | f(x_t), \text{diag}(g(x_t)))$$



Denoising diffusion model



$$\log p_{\theta}(x_0)$$

$$= \log E_{q(X_{1:T}|x_0)} \left[\frac{p(X_0, X_{1:T})}{q(X_{1:T}|x_0)} \right]$$

$$\geq E_{q(X_{1:T}|X_0)} \left[\log \frac{p(X_0, X_{1:T})}{q(X_{1:T}|X_0)} \right]$$

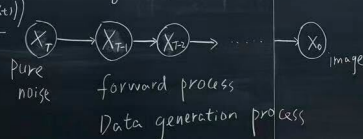
$$= E_{q(X_{1:T}|X_0)} [\log p(X_0, X_{1:T}) - \log q(X_{1:T}|X_0)]$$

$$\frac{p(x_i) \prod_{t=1}^T p(x_{t+1} | x_t)}{N(0, I)}$$

$$N(x_{t-1} | f(x_t), \text{diag}(g(x_t)))$$



Denoising diffusion models



models

image

process

$$q(X_{1:T} | X_0) = \prod_{t=1}^T q(X_t | X_{t-1})$$

$$q(X_0, X_1, \dots, X_T) = q(X_0) \prod_{t=1}^T q(X_t | X_{t-1})$$

$$q(X_t | X_{t-1}) = \mathcal{N}(X_t; \sqrt{1 - \beta_t} X_{t-1}, \beta_t I)$$

$0 < \beta_t < 1$ noise schedule

Empirical $q(X_0) = \frac{1}{N}$ distribution

$$q(X_t | X_0) = \mathcal{N}(X_t; \sqrt{\alpha_t} X_0, (1 - \alpha_t) I)$$

$\alpha_t = \prod_{s=1}^t (1 - \beta_s)$

Composition of Gaussians

Generate data

Latent diffusion model

$\hat{X}_T \sim \mathcal{N}(0, I)$

$\hat{X}_{T-1} \sim \mathcal{N}(X_{T-1} | \hat{X}_T)$

$\hat{X}_0 \sim \mathcal{N}(X_0 | \hat{X}_1)$

image

$0 \ 0 \ 0 \ 0 \ 0$

X_T (10,000) → VAE Encoder → [] (100) → [] (100) → [] (100) → [] (100) → VAE Decoder → X_0 (10,000)

θ^*

X_S

score-based diffusion models

Backward process

$$q(x_t|x_{t+1}) = N(x_t, \sqrt{1-\beta_t} \sigma_t^2)$$

$$x_t = \sqrt{1-\beta_t} x_{t+1} + \sqrt{\beta_t} \epsilon_t$$

$$x_t = \sqrt{1-\beta_t} x_{t+1} + \sqrt{\beta_t} \epsilon_t$$

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$$x_t = \sqrt{1-\beta_t} x_{t+1} + \sqrt{\beta_t} \epsilon_t$$

at $t \rightarrow 0$

$$dx_t = -\frac{1}{2} \beta_t^{-1} \sigma_t^2 dt + \sqrt{\beta_t} dW_t$$

Forward process

$$dx_t = \left[-\frac{1}{2} \beta_t^{-1} \sigma_t^2 + \beta_t^{-1} \log \frac{q(x_t|x_{t+1})}{q(x_{t+1}|x_t)} \right] dt + \sqrt{\beta_t} dW_t$$

stochastic differential equation (SDE)

x_t

$$u_t = N(0, \sigma_t^2)$$

score function

$$\beta_t$$

$$\sigma_t$$

$$N(0, \sigma_t^2)$$

$$\min_{\theta} \mathbb{E}_{t \sim U(0,T)} \mathbb{E}_{x_t \sim q(x_t|x_0)} \left\| \frac{\partial}{\partial x_t} \log q(x_t|x_0) \right\|_2^2$$

$$\approx \mathbb{E}_{t \sim U(0,T)} \mathbb{E}_{x_t \sim q(x_t|x_0)} \left\| \frac{\partial}{\partial x_t} \log q(x_t|x_0) \right\|_2^2$$

$$N(x_0, \beta_t x_0, \sigma_t^2 \mathbb{I})$$

$$\beta_t = e^{-\int_0^t \beta_s ds}$$

$$\sigma_t^2 = 1 - e^{-\int_0^t \beta_s ds}$$